Frequently Asked Questions

What are these handouts?

From 2017 to 2020, I tutored a few high school students per year, and wrote a set of common homework assignments. From 2021 to 2024, I trained the US Physics Team, while gradually refining the assignments into a comprehensive set of handouts. They contain over 250 worked examples and 1,000 varied and challenging problems. Many USAPhO, IPhO, APhO, and EuPhO problems are included, but I've also taken problems from more obscure competitions, recent research, the Victorian-era Cambridge Tripos, graduate qualifying exams, and more. I've also written some problems myself, to illustrate ideas that common textbooks don't explain well, extended classic problems to bring out subtle points, corrected competition questions when needed, and included in-depth discussions of the context for problems from history or more advanced physics.

The core USAPhO material is covered in 24 handouts. There are also 3 review handouts, and 6 more advanced handouts. Since Olympiad physics ultimately is just real physics with trickier problems and elementary math, you could probably also use the handouts to learn university physics; they cover approximately the first half of an undergraduate degree.

These handouts were greatly improved with the help of many sharp readers. In particular, I thank Gopal Goel and Sean Chen for helping me write the original solutions, and Physics Hub, Stefan Ivanov, and Annie Wang for improving the diagrams. I thank Edward Chen, Alex Huang, Agastya Goel, Stefan Ivanov, Miles Zhang, and Kai Wen Teo for spotting many errors and oversights, and William Guan, Brian Zhang, and Jerry Liu for suggesting interesting followup questions.

Who has used the handouts?

From 2017 to 2020, I tutored 11 students using these handouts. 10 won USAPhO gold medals or equivalent, 6 qualified for USAPhO camp, and 3 won IPhO gold medals. I trained the US traveling team for the 2021 IPhO and 2024 EuPhO primarily using these handouts, and both years the team achieved a perfect 5 gold medal finish. Currently, the handouts are used by students in countries around the world to prepare for the IPhO.

Do you offer tutoring now?

No, I'm too busy. But the handouts are designed so students can follow them on their own, and many have done just that. You can meet students working through them on this Discord server.

Why are you giving these away for free?

When I was in high school, I heard about the Physics Olympiad from my physics teacher, spent an afternoon searching online for a book recommendation, and bugged my parents to buy Halliday, Resnick, and Krane. They got me a $20 used copy with the spine falling off, and I spent an enjoyable year puzzling over it by myself, reading it on the weekends and during chemistry class. When I went to the IPhO that summer, I still only had that one book, held together with layers of tape.

This shows that Olympiads are one of the most accessible high school activities in the world. If you want to be a chess prodigy, a musician, or an elite athlete, you need expensive coaches, and parents who will drive you to practice and fly you to competitions around the world. If you want to do robotics, you need to go to a high school whose club has a five-figure budget, and if you want to

do research, you'll almost always be doing it in a university lab with a multimillion dollar budget. But to learn physics, you only need one book and your own mind.

On the other hand, while there was a path, it wasn't well-paved. I spent hours at a time stuck on little things, because of missing steps in solutions, ambiguous statements, or outright errors. I didn't know much beyond the textbook, which meant I had almost no knowledge of subjects like fluid dynamics and circuits, and online explanations were of dubious quality. I didn't have a broad range of sources for training problems, or any idea of how to compare their difficulty, so I was blindsided by IPhO questions that were different from what I'd seen before. Nobody told me the big picture, the deep links between all the fields of physics I was learning; I pieced some of it together myself, but the process was unreliable and slow.

My goal with these handouts is to shorten the way for future students, both within the United States and around the world. In the original spirit of the Olympiads, I hope they spark interest in physics for years to come.

How can I make similar looking documents?

You can download a TeX template for the handouts here. I use Sublime Text with LaTeXTools, though any common LaTeX editor will work well.

How do I know if I'm ready to start using the handouts?

Before starting, you should have completed most of a standard calculus-based introductory physics book, such as Halliday, Resnick, and Krane. (For more guidance, see my advice on learning introductory physics.) You should also have some problem solving experience from entry-level competitions. To judge this, spend a few days on the basic problems in the preliminary problem set. For the handouts to be most useful, you should be able to solve over 75% of them.

What if I can only do the mechanics preliminary problems?

I recommend learning at least introductory mechanics and electromagnetism before starting the handouts. The reason is that a good calculus-based introductory textbook gradually increases the depth of its mathematics and the sophistication of its physical arguments. In the early mechanics chapters, it'll explain in great detail what it means to take an integral or a cross product, or to consider a system or a differential element, but by the late electromagnetism chapters these operations are used freely, and expected to be second nature. My handouts also gradually increase in level, but they start where a typical introductory textbook ends. So if you go straight from textbook mechanics to the mechanics handouts, you might face a tough difficulty jump.

What if I've aced both the AP Physics C exams?

The average AP Physics problem requires recalling basic formulas, plugging numbers into them, or at most combining 2-3 standard formulas and solving for a variable. By contrast, in Olympiad problems you are often expected to come up with formulas you haven't seen before, by thinking carefully about the physical principles governing the situation, and to fluently apply them. Furthermore, the diversity of Olympiad questions is at least 10 times greater.

So the answer depends on how well you've personally learned physics. Again, the best way to check is to try some old exams, or the preliminary problem set. If you find them too challenging, you can review a good introductory physics textbook, focusing on its harder questions.

What if I've done Halliday and Resnick, but still find the handouts very difficult?

There's definitely a difficulty jump, but keep in mind that handout questions should take longer on average. If you want another stepping stone, try questions from an easier competition. When I was in high school, I did this by working through pre-2007 $F = ma$ and USAPhO exams. Today, you could also try Morin's *Problems and Solutions in Introductory Mechanics*, or the book *200 Estonian Physics Olympiad Problems*. These resources vary widely in style, but all of them can help you get used to longer questions, so choose whichever you like best.

How much math background do I need?

Only what's necessary to understand Halliday, Resnick, and Krane, i.e. calculus and occasionally some multivariable calculus. You don't need to know any Olympiad math.

How long should it take to go through the handouts?

It varies a lot. On average, students took about 10 months to go through the core 24 handouts. The amount of time they worked varied from 8 to 15 hours per week. It also varied between handouts: sometimes a student would blaze through an early one in 2 days, but then take 2 weeks on a later one because more background reading was required, or because they wanted to stop and think about the material more deeply. (The handouts contain lots of references to textbooks and papers, in case you want to take the scenic route.) I wouldn't worry about how much time things are taking you, as long as you're learning new things and enjoying yourself.

By the way, my metric for completing a handout is solving about 2/3 of the problems, weighted by points. For most purposes, doing all of every handout is overkill, and you can pick and choose what looks most interesting to you. Also, there's no need to spend a long time trying to solve the examples on your own – they are intended to introduce new ideas, and sometimes they're extremely tricky. The example solutions are designed to be read as you go.

But you should absolutely *not* just read all the problem solutions, one after the other. I've tried to make them clear, so they'll "make sense" as you read them, but if you haven't actually tried a problem, you won't understand how to come up with its solution independently, or how to distinguish the right solution from similar-sounding wrong solutions. When students blast through all the handouts this way, they end up very confused. They only remember shallow information like "use trick X for problem Y in handout Z", which misses the point, makes the problem useless for practice later, and doesn't generalize to unseen problems.

How did you choose the problems?

I've tried to choose each problem to teach a distinct physical lesson, which means:

- Few "cookie-cutter" questions. I try to give one or two problems or examples for each new technique you should know, but that's enough. If you want more routine practice problems, you can easily find them in standard books. But it really doesn't make sense to demand, say, a dozen identical problems about massless pulleys. In Olympiads, success comes from deep understanding gleaned from a few tough problems, not just mindlessly exercising your fingers on many easy ones. Each problem should teach a lesson not present in any other.
- Few "hidden math" questions, i.e. questions where the physics is just used as a thin shell for a math problem. For example, ideal gas heat engine problems are pretty routine, so why not

calculate the efficiency of an elephant-shaped cycle? Or maybe reconstruct the axes in a PV diagram with ruler and compass? Pulley problems are pretty easy, so why not make them harder by having six pulleys, so students have to solve a giant system of equations, or infinitely many pulleys, so that students have to sum infinite series? Resistor networks are straightforward, so why not build fractals out of them?

But these contrived questions all have trivial *physical* content: it's always completely obvious what physics equations should be used, and the difficulty comes only from solving those equations. Worse, none of these things resemble anything a real physicist would ever encounter. There actually are plenty of interesting questions involving ideal gases, pulleys, and resistors, but you can't get to them by starting with trivial questions and stacking on mathematical complications.

- Few "one weird trick" questions. A lot of easier Olympiad problems are just cookie cutter problems that also require a single key idea. As a very simple example, in many problems that involve objects initially arranged at the vertices of a regular polygon, you can use symmetry to conclude that this always remains true. These ideas can be quite beautiful, and they're often important in more advanced physics. But I think it's sufficient to see each one just once or twice, in the simplest possible context. You don't need twenty examples to see the pattern.

- No "two weird trick" questions. Sometimes people construct tough problems by pasting together simpler ones, but I don't think that's a lot of fun, because the whole doesn't end up greater than the sum of the parts. A great tough problem should contain an *irreducible* nugget of insight: a completely new idea that is already as simple as it can possibly be, yet still difficult to find. In other words, I constructed the handouts to be comprehensive, but also as easy as possible![1]

You might be wondering: if you can't make great problems by tweaking existing problems, adding tricks to them, or combining existing tricks, then how are they created at all? Usually, they are produced by subtraction, not addition. A physicist spots something interesting, such as a setup commonly used in their research area, an insight from an advanced course, a recent paper, a piece of technology, or even a children's toy. Then they think hard about how to present it in the simplest possible way, so that it can be solved using high school knowledge alone. The benefit of this kind of question is that when you solve them, you're learning something real, whether it's a preview of advanced material, some practical know-how, or an insight into the physics of everyday life.

Why is handout problem X very similar to a problem on a recent exam?

The handouts are continuously updated, and I incorporate recent exam problems if they demonstrate an interesting new lesson. But about half the time, the handout problem was already there before the exam; this reflects the fact that there's a finite number of interesting physics Olympiad problems.

Are these handouts enough to get USAPhO or IPhO gold medals?

No amount of reading can guarantee any outcome. That said, the theoretical coverage of the handouts is extensive; if you finish them, I think further Olympiad theory practice has diminishing returns. Afterward, I recommend moving on to experimental practice and undergraduate physics.

[1] Physics is about attaining a clearer view of the world, and nature doesn't award points for difficulty. I want students to reach the view at the top by climbing the most *natural* path up the mountain, not the most painful one.

What if I just want a USAPhO bronze medal or lower?

If that's your main goal, it would be better to understand introductory physics more deeply, and to practice on simpler questions. See my advice on learning introductory physics for some options.

Do these handouts cover enough for my country's IPhO qualification exam?

The handouts complement and extend the introductory physics curriculum in English-speaking countries, and from these countries there are dozens of success stories. In former Soviet countries (and India) the traditional curriculum has less relativity and waves but more optics, less electromagnetism but more circuits, and less calculus but more geometric tricks. I've tried to cover all topics, but you'll probably need extra practice, most notably on optics. In countries with less developed Olympiad systems, the handouts will be harder than the selection exams, but the exams themselves might be idiosyncratic. (I've seen selection exams use questions with ill-defined answers, or ask trivia like the number of watts in one horsepower.) Finally, East Asian countries such as China and South Korea have many strong competitors, distinguished at the top end by test taking speed and reliability, and knowledge of advanced physics. For students from these countries, the handouts can work as a supplemental source of interesting problems, but probably not as a main resource.

Where can I find exam files and solutions?

Almost all relevant exams are on phoXiv, and links will be provided for exceptions.

What if I'm stuck on a problem?

Try reading a line of the solution and trying to continue from there. If you're confused about a basic problem, try reviewing the background reading. You can try emailing me if you have a conceptual question which isn't already answered in the solutions, but I might be too busy for a timely reply.

What if I find an error in the solutions?

For solutions in the handouts themselves, I always welcome error reports. Learning new things is hard enough without typos, and if you point one out, you'll help clear the way for many future students. Before reporting an error, make sure you're using the latest version of the handouts here.

I would also be very interested if you spot an error in a USAPhO solution between 2008 and 2024, since I've been working with AAPT to improve those solutions for the past few years. Before reporting an error, please make sure you're looking at the latest versions, available here. I'm not involved with exams after 2024, so direct inquiries about those to AAPT.

If you find that my handouts refer to a flawed IPhO or APhO question, without a warning, I would appreciate knowing! You should also get in touch with Stefan Ivanov, who maintains a combined errata list here.